

Structural Analysis of High-Energy Physics Citation Network Through Community Detection

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Complex Network Project Report
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This project tries to perform a structural analysis of the high-energy citation network, from SNAP database through community detection. The hypothesis of this project is that the communities are naturally separated by sub-fields. To test this hypothesis, first metadata has been generated by labeling each paper by up to three sub-fields. Then, the communities were detected with the infomap algorithm, which minimizes the description length. Lastly, to test the main hypothesis of this project, a Fisher's exact test has been conducted. The results confirm our hypothesis and show a very low P-value for the sub-fields in each community.

Keywords: Complex Network, Network analysis, Cluster Analysis, Community Detection, Citation Network

1. Data

The data for this project has been acquired from Stanford Large Network Dataset Collection-SNAP.[1] The data is the citation network of papers in the arXiv labeled with hep-ph (Physics High-energy Phenomenology) and contains all the papers in the period from January 1993 to April 2003 (124 months). Each paper has the identifier of the arXiv id. This dataset contains 34,546 papers with 421,578 edges. If a paper i cites paper j , the graph contains a directed edge from i to j . If a paper cites or is cited by a paper outside the dataset, the graph does not contain any information about this. The data also contains a separate file with the publishing time of the papers, which was not used for this project.

■ 2. Methods

The main hypothesis of this project was that the communities in the citation graph are naturally separated with respect to their sub-fields. The first step for showing this objective was to label each paper with its proper sub-fields. To achieve this, titles and abstracts of all papers have been extracted. Then, a simple yet comprehensive dictionary and a word search have been applied to label the papers. Then the infomap algorithm has been used to detect communities. Lastly, Fisher's exact test has been applied to report the accuracy of the communities detected and to test how well separated were the detected communities. The final communities were then visualized.

Python programming language has been used for this project. All required dependencies and libraries, alongside the scripts, have been provided in a public GitHub repository.[2]

■ 2.1 Metadata and Labeling

To test the main hypothesis of this project, it was necessary to have sub-field tags for each paper. The main data did not have any information on the sub-fields of each paper. All papers in this dataset were labeled by high-energy phenomenology field tag in arXiv, but no more information on the sub-fields was provided. So labeling the papers was an important first step of this project.

To achieve this, two separate files have been generated. The first metadata file contains titles and abstracts. This file has been generated with a Python function named "fetch-metadata", which takes the paper id and returns the abstract and title for the corresponding paper by sending and fetching a request from arXiv and storing it in a JSON file.

After having all the abstracts and titles, the labeling assignment has been carried out. This task was done by first creating a dictionary of subfields. This dictionary contains a total of 15 sub-fields and 116 phrases. Then a phrase search was conducted on abstracts and titles and assigned up to three labels per paper, and an additional "unknown" label in case of a no-match result.

■ 2.2 Community Detection

The infomap algorithm seemed suitable for conducting the community detection task, as the flow of information plays a significant role in the nature of the citation network. This algorithm is based on minimizing the description length of a random walker through the network, such that a random walker tends to stay in each community for long stretches of time. By focusing on flow (i.e., how often a random walker would traverse the connections between nodes), Infomap naturally captures the community structure based on actual data flows, making it

suitable for networks with clear flow dynamics.

The description length that the infomap algorithm seeks to minimize quantifies the information needed to describe a random walker's movements. The description length can be divided into two parts: index code length, and module code length. The index code length measures the information needed to describe the transitions between different communities (modules). It represents the overhead of the walker moving from one community to another and is determined by the frequency of these transitions. The probability of moving between communities dictates how much information (in bits) is needed to encode these movements. If transitions between certain communities are frequent, those transitions can be encoded with fewer bits. The module code length describes the movements within communities. Within each community, the frequency of movement between nodes dictates the encoding; more frequent paths within a community require fewer bits to describe. This algorithm has three steps: 1. Initialization: Start with each node in its own community. 2. Aggregation: Iteratively group nodes together into communities such that each merger results in the greatest decrease in the description length of the random walker's movements. 3. Refinement: Once no straightforward mergers produce a decrease, the algorithm can fine-tune by moving individual nodes between communities or by recursively applying the entire process to communities themselves.[3] The infomap library has been used to incorporate this algorithm into our project.

■ 2.3 Fisher's Exact Test

After the communities had been detected, Fisher's exact test was carried out to test the main hypothesis of this project, and to test how effective and well-separated the communities were with respect to the subfields within each community. Fisher's exact test is a statistical test to determine whether there exists a nonrandom association between two variables. The first calculated index in this project is the odds-ratio(OR). The odds ratio measures the odds of an event occurring in one group to the odds of it occurring in another group. The other measured index is the P-value. P-value provides a measure of the significance of that association, quantifying whether the observed association could be due to chance, whereas odds-ratio provides a measure of the strength of the association or non-independence between two data values, and is a descriptive measure. Both metrics are essential to fully understand the results of Fisher's exact test: the odds ratio contextualizes the nature of the association, while the p-value tells us about the reliability of this observation. Consider the following contingency table:

	sub-field present	sub-field absent
in community	a	b
out community	c	d

Each index in this table represents the count of occurrences. a: Number of papers in the community with a specific subfield; b: Number of papers in the community without the subfield; c: Number of papers outside the community with the subfield; d: Number of papers outside the community without the subfield. Then the odds ratio is:

$$OR = \frac{ad}{bc} \quad (1)$$

A low p-value (typically ≤ 0.05) indicates that the observed distribution of data is unlikely under the null hypothesis, leading to the rejection of the null hypothesis.

There are two steps to compute p-value. The first step is the probability of observing this specific table under the null-hypothesis (the default or baseline assumption that there is no effect or no relationship between two measured phenomena). This probability is as follows:

$$P(a, b, c, d) = \frac{\binom{a+c}{a} \binom{b+d}{b}}{\binom{N}{a+b}} \quad (2)$$

where $N = a + b + c + d$, the total number of papers. The second step is then to compute the sum probabilities of all tables as extreme as the observed table (with the same marginal values). The P-value then is calculated by:

$$\text{P-value} = \sum_{\text{all } a' \text{ as extreme as the table}} P(a', b', c', d') \quad (3)$$

Typically, "as extreme" is assessed by looking at tables with an odds ratio that is as far or farther from 1 than the odds ratio of the observed table. A very low p-value (typically less than 0.05) leads to the rejection of the null hypothesis, suggesting that the observed effect or relationship is statistically significant.[4] To perform Fisher's exact test and obtain the odds ratio and P-value, the `scipy.stats` python library has been used.

■ 2.4 Visualization

To visualize the citation network, few considerations have been made. First the nodes have been visualized by sizes proportional to their degree. Second, a random color has been assigned to each community.

Third, a Kamada-Kawai layout has been chosen for better node separation. Lastly, only the nodes with a higher degree than 60 have been visualized.

3. Results

Firstly, some network metrics have been computed for understanding both the overall and internal structure. Global Edge Density (0.000353) is relatively low, indicating that the network is sparse. In academic networks, this sparsity is typical as most papers cite only a relevant subset of other papers. Global Clustering Coefficient (0.143) indicates a moderate level of clustering, suggesting that nodes (papers) tend to cluster around common neighbors more than would be expected in a random network. This could reflect common citation patterns within specific fields or topics. Global Average Degree Centrality (0.000707) is also low, reflecting the sparsity but also indicating that on average, nodes are not highly connected, which is typical in large citation networks where each paper cites only a limited number of other papers. Global Average Betweenness Centrality (0.000126) is a low that value suggests that there are few nodes that act as critical points of control within the network. Most papers do not serve as pivotal bridges between other papers.

The infomap algorithm has detected 66 communities, with 6 major communities meaning they had more than 500 papers. The remaining minor communities each had less than 10 papers. Since six major communities have been detected, if we reduced the sub-fields to six more general areas, probably each would have a high odds ratio on one community. However, this has not been tested and was not part of the project's objective, and is just a hypothesis.

Measuring metrics in communities can be categorized as: In the network analysis(see the table), Communities 3, 4, 5, and 6 show strong internal connections and group cohesion, typical of well-established or highly collaborative groups. Community 1, while larger and interacting more than average, behaves similarly to the broader network without standout central figures. Community 2, despite high internal bonding, lacks key central nodes, suggesting a flat structure where many nodes are equally important. Community 3 and Community 4 are particularly cohesive, with Community 4 showing the highest closeness within groups, though it interacts less outside its own group. Community 5 appears to be a niche or emerging area, strongly connected internally but less influential outside.

Community	Most/ Least Represented Topic	Odds Ratio	P-value	Notes on metrics
1	Supersymmetry/ Phenomenology of QCD	15.23/ 0.01	0.0000/ 0.0000	Average centrality, cohesive but not central in control.
2	Experimental QCD/ Cosmology	35.06/ 0.02	0.0000/ 0.0000	Lower centrality, suggesting decentralized structure.
3	QCD/ cosmology	2.49/ 0.04	0.0000/ 0.0000	Slightly higher centrality, more reliance on central nodes.
4	Phenomenology of QCD/ Experimental QCD	0.28/ 0.08	0.0000/ 0.0000	Highest clustering, insular interactions within community.
5	QCD/ Supersymetry	3.85/ 0.02	0.0000/ 0.0000	Emerging or specialized field with strong internal ties.
6	Phenomenology of QCD and Unknown/ (not meaningful)	6.35 and 4.92 /not meaningful	0.0000 and 0.0000/ -	Extremely high connectivity, evenly shared knowledge, very low centrality. Strange metrics due to the strong presence of unknown.

Community 6 stands out with the highest connections and group closeness but very low centrality, this might be due to the strong presence of the unknown field among this small community. Also the least represented topic in this community has not been mentioned in the table because of the very low number of papers on this topic. The lower external influence of several communities despite their strong internal ties suggests these might be specialized or developing areas gaining traction internally but not yet widely influential. Variability in centrality across communities highlights different roles nodes play, with higher values pointing to influential nodes crucial for spreading information within their groups.

The Fisher's exact test results, the sub-fields distribution, the global as well as metrics for communities, and the dominant sub-fields in each community were reported as follows:

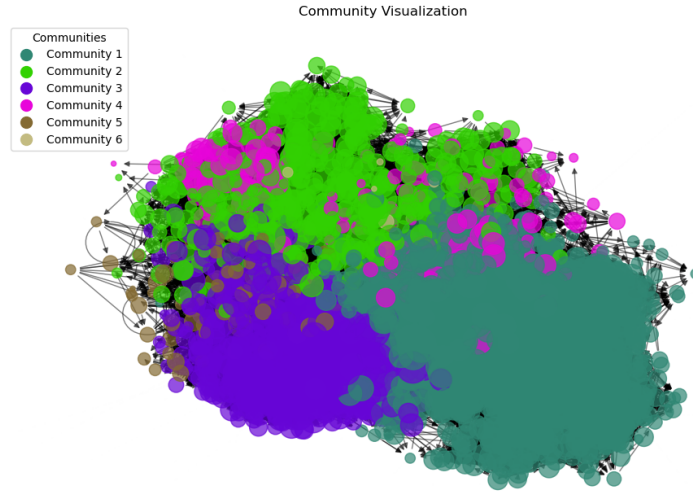


Figure 1. The visualization of citation network of high-energy physics, phenomenology from January 1993 to April 2003 from arXiv. The major communities detected by this project have been assigned random colors. Only nodes with a higher degree than 60 have been visualized. The node sizes are proportional to their degree.

Global Metrics:

global_edge_density: 0.00035
 global_clustering_coefficient: 0.14326
 global_avg_degree_centrality: 0.00071
 global_avg_betweenness_centrality: 0.00013

Community 1:

edge_density: 0.0009189744308478654
 avg_clustering: 0.14790413620879486
 avg_degree_centrality: 0.0008097969479781596
 avg_betweenness_centrality: 0.00012522290464378336
 Subfield Distribution:
 Gravitational Physics: 600 - Odds Ratio: 4.38, P-value: 0.0000
 Beyond the Standard Model: 1690 - Odds Ratio: 3.33, P-value: 0.0000
 Particle Physics: 2429 - Odds Ratio: 1.60, P-value: 0.0000
 String Theory: 716 - Odds Ratio: 14.40, P-value: 0.0000
 Electroweak Physics: 5586 - Odds Ratio: 2.58, P-value: 0.0000
 Cosmology: 1487 - Odds Ratio: 3.70, P-value: 0.0000
 Quantum Field Theory: 862 - Odds Ratio: 0.53, P-value: 0.0000
 Supersymmetry: 4091 - Odds Ratio: 15.23, P-value: 0.0000
 Quantum Chromodynamics: 3124 - Odds Ratio: 0.15, P-value: 0.0000
 Neutrino Physics: 5110 - Odds Ratio: 9.61, P-value: 0.0000
 General Theoretical Physics: 1207 - Odds Ratio: 1.99, P-value: 0.0000
 Experimental QCD: 48 - Odds Ratio: 0.09, P-value: 0.0000
 Unknown: 1035 - Odds Ratio: 0.47, P-value: 0.0000

Quantum Electrodynamics: 335 - Odds Ratio: 0.94, P-value: 0.3744
 Non-Perturbative QCD: 35 - Odds Ratio: 0.06, P-value: 0.0000
 Perturbative QCD: 59 - Odds Ratio: 0.07, P-value: 0.0000
 Phenomenology of QCD: 2 - Odds Ratio: 0.01, P-value: 0.0000

Community 2:

edge_density: 0.0012614396856023464
 avg_clustering: 0.16511371223650725
 avg_degree_centrality: 0.0006462561867260277
 avg_betweenness_centrality: 8.336907832871411e-05
 Subfield Distribution:
 Quantum Chromodynamics: 5860 - Odds Ratio: 4.08, P-value: 0.0000
 Experimental QCD: 710 - Odds Ratio: 35.06, P-value: 0.0000
 Neutrino Physics: 590 - Odds Ratio: 0.29, P-value: 0.0000
 Quantum Field Theory: 594 - Odds Ratio: 0.78, P-value: 0.0000
 Unknown: 1031 - Odds Ratio: 1.19, P-value: 0.0000
 Particle Physics: 1319 - Odds Ratio: 1.32, P-value: 0.0000
 Perturbative QCD: 745 - Odds Ratio: 6.81, P-value: 0.0000
 Electroweak Physics: 1317 - Odds Ratio: 0.43, P-value: 0.0000
 Phenomenology of QCD: 207 - Odds Ratio: 4.38, P-value: 0.0000
 Non-Perturbative QCD: 201 - Odds Ratio: 1.05, P-value: 0.5896
 Gravitational Physics: 12 - Odds Ratio: 0.05, P-value: 0.0000
 Cosmology: 17 - Odds Ratio: 0.02, P-value: 0.0000
 Beyond the Standard Model: 75 - Odds Ratio: 0.09, P-value: 0.0000
 Supersymmetry: 72 - Odds Ratio: 0.04, P-value: 0.0000
 General Theoretical Physics: 239 - Odds Ratio: 0.40, P-value: 0.0000
 Quantum Electrodynamics: 188 - Odds Ratio: 0.94, P-value: 0.5089
 String Theory: 16 - Odds Ratio: 0.07, P-value: 0.0000

Community 3:

edge_density: 0.00268586126656957
 avg_clustering: 0.1577633576608313
 avg_degree_centrality: 0.0008427270744909596
 avg_betweenness_centrality: 0.00019113706305873683
 Subfield Distribution:
 Quantum Chromodynamics: 3103 - Odds Ratio: 2.49, P-value: 0.0000
 Neutrino Physics: 319 - Odds Ratio: 0.30, P-value: 0.0000
 Electroweak Physics: 1331 - Odds Ratio: 1.03, P-value: 0.3532
 Quantum Field Theory: 394 - Odds Ratio: 0.95, P-value: 0.3452
 Beyond the Standard Model: 462 - Odds Ratio: 1.51, P-value: 0.0000
 Supersymmetry: 251 - Odds Ratio: 0.33, P-value: 0.0000
 Perturbative QCD: 194 - Odds Ratio: 1.32, P-value: 0.0007
 Particle Physics: 483 - Odds Ratio: 0.69, P-value: 0.0000
 Non-Perturbative QCD: 193 - Odds Ratio: 1.97, P-value: 0.0000
 Unknown: 559 - Odds Ratio: 1.06, P-value: 0.2070
 Gravitational Physics: 17 - Odds Ratio: 0.13, P-value: 0.0000
 General Theoretical Physics: 123 - Odds Ratio: 0.38, P-value: 0.0000
 Experimental QCD: 7 - Odds Ratio: 0.06, P-value: 0.0000
 Quantum Electrodynamics: 103 - Odds Ratio: 0.89, P-value: 0.2839
 Cosmology: 15 - Odds Ratio: 0.04, P-value: 0.0000
 String Theory: 20 - Odds Ratio: 0.16, P-value: 0.0000
 Phenomenology of QCD: 3 - Odds Ratio: 0.05, P-value: 0.0000

Community 4:

edge_density: 0.0021909070119579058
 avg_clustering: 0.18377007668511813
 avg_degree_centrality: 0.0005442101584014067
 avg_betweenness_centrality: 0.00013265347798945083
 Subfield Distribution:
 Cosmology: 551 - Odds Ratio: 3.16, P-value: 0.0000
 Electroweak Physics: 983 - Odds Ratio: 0.89, P-value: 0.0019
 Quantum Chromodynamics: 1643 - Odds Ratio: 0.84, P-value: 0.0000
 Unknown: 636 - Odds Ratio: 1.66, P-value: 0.0000
 Supersymmetry: 175 - Odds Ratio: 0.28, P-value: 0.0000
 Particle Physics: 314 - Odds Ratio: 0.53, P-value: 0.0000
 Quantum Field Theory: 703 - Odds Ratio: 2.75, P-value: 0.0000
 Beyond the Standard Model: 135 - Odds Ratio: 0.44, P-value: 0.0000

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General Theoretical Physics: 317 - Odds Ratio: 1.46, P-value: 0.0000
Neutrino Physics: 100 - Odds Ratio: 0.11, P-value: 0.0000
String Theory: 31 - Odds Ratio: 0.32, P-value: 0.0000
Gravitational Physics: 152 - Odds Ratio: 1.91, P-value: 0.0000
Non-Perturbative QCD: 201 - Odds Ratio: 2.65, P-value: 0.0000
Quantum Electrodynamics: 161 - Odds Ratio: 1.95, P-value: 0.0000
Phenomenology of QCD: 112 - Odds Ratio: 3.63, P-value: 0.0000
Perturbative QCD: 46 - Odds Ratio: 0.33, P-value: 0.0000
Experimental QCD: 8 - Odds Ratio: 0.08, P-value: 0.0000

Community 5:
edge_density: 0.002392181013161233
avg_clustering: 0.16652403087222173
avg_degree_centrality: 0.000465849643776555
avg_betweenness_centrality: 0.00016530346210836606
Subfield Distribution:
General Theoretical Physics: 200 - Odds Ratio: 1.25, P-value: 0.0051
Electroweak Physics: 285 - Odds Ratio: 0.28, P-value: 0.0000
Quantum Chromodynamics: 2054 - Odds Ratio: 3.85, P-value: 0.0000
Unknown: 391 - Odds Ratio: 1.35, P-value: 0.0000
Quantum Field Theory: 442 - Odds Ratio: 2.20, P-value: 0.0000
Beyond the Standard Model: 93 - Odds Ratio: 0.44, P-value: 0.0000
Neutrino Physics: 60 - Odds Ratio: 0.09, P-value: 0.0000
Perturbative QCD: 41 - Odds Ratio: 0.43, P-value: 0.0000
Non-Perturbative QCD: 179 - Odds Ratio: 3.38, P-value: 0.0000
Quantum Electrodynamics: 41 - Odds Ratio: 0.60, P-value: 0.0007
Cosmology: 13 - Odds Ratio: 0.07, P-value: 0.0000
Particle Physics: 62 - Odds Ratio: 0.14, P-value: 0.0000
Gravitational Physics: 7 - Odds Ratio: 0.10, P-value: 0.0000
Supersymmetry: 10 - Odds Ratio: 0.02, P-value: 0.0000
Experimental QCD: 6 - Odds Ratio: 0.09, P-value: 0.0000
Phenomenology of QCD: 5 - Odds Ratio: 0.16, P-value: 0.0000
String Theory: 5 - Odds Ratio: 0.07, P-value: 0.0000

Community 6:
edge_density: 0.006660952476989182
avg_clustering: 0.19195129528456814
avg_degree_centrality: 0.00026871062948768775
avg_betweenness_centrality: 1.6974794341992025e-05
Subfield Distribution:
Perturbative QCD: 45 - Odds Ratio: 2.46, P-value: 0.0000
Quantum Chromodynamics: 288 - Odds Ratio: 1.06, P-value: 0.4822
Particle Physics: 85 - Odds Ratio: 1.05, P-value: 0.6735
Unknown: 224 - Odds Ratio: 4.92, P-value: 0.0000
Electroweak Physics: 24 - Odds Ratio: 0.11, P-value: 0.0000
Quantum Field Theory: 9 - Odds Ratio: 0.16, P-value: 0.0000
Non-Perturbative QCD: 7 - Odds Ratio: 0.48, P-value: 0.0567
Neutrino Physics: 12 - Odds Ratio: 0.09, P-value: 0.0000
Quantum Electrodynamics: 2 - Odds Ratio: 0.13, P-value: 0.0001
Phenomenology of QCD: 35 - Odds Ratio: 6.35, P-value: 0.0000
Beyond the Standard Model: 1 - Odds Ratio: 0.02, P-value: 0.0000
Supersymmetry: 1 - Odds Ratio: 0.01, P-value: 0.0000
Experimental QCD: 3 - Odds Ratio: 0.21, P-value: 0.0012
Cosmology: 1 - Odds Ratio: 0.03, P-value: 0.0000

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The results of these communities confirm our main hypothesis, that the communities were naturally separated in correspond to their sub-fields, as we have meaningfully low p-values for almost all sub-fields in all our communities. The visualization of the network can also be seen in figure-1, with the mentioned considerations in the methods section.

4. Supplemental Information

The citation dataset for this project has been obtained from SNAP.[1]. All the scripts, and results have been provided in a public gitHub repository.[2]

References

- [1] <https://snap.stanford.edu/data/cit-HepPh.html>
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